TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	81.4 ± 2.91
Correlation Stability	71.7 ± 0.46
Distribution Consistency	85.2 ± 3.37
Entropy Preservation	63.2 ± 2.40
Mean Z Score Drift	0.2 ± 0.07
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	100.0 ± 0.00
Feature Accuracy	73.8 ± 7.43
Novelty Score	0.9 ± 0.01

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Dataset Audit: chronic_kidney_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	400
Columns	25
Numeric features	25
Categorical features	0
Missing cells (%)	10.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	51.483	66.881	+15.398	29.9%	Drift
bp	76.469	79.562	+3.093	4.0%	OK
sg	1.017	1.020	+0.002	0.2%	OK
al	1.017	1.222	+0.205	20.2%	Warn
su	0.450	0.625	+0.175	38.8%	Drift
rbc	0.190	0.156	-0.033	17.6%	Warn
pc	0.227	0.266	+0.039	17.1%	Warn
pcc	0.106	0.138	+0.031	29.6%	Drift
ba	0.056	0.072	+0.016	29.4%	Drift
bgr	148.037	182.156	+34.120	23.0%	Warn
bu	57.426	86.695	+29.269	51.0%	Drift
sc	3.072	4.730	+1.657	53.9%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	81.4 ± 2.91
Correlation Stability (*2)	71.7 ± 0.46
Distribution Consistency (*3)	85.2 ± 3.37
Entropy Preservation (*4)	63.2 ± 2.40
Mean Z Score Drift (*5)	0.2 ± 0.07
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	100.0 ± 0.00
Feature Accuracy (*10)	73.8 ± 7.43
Novelty Score (*11)	0.9 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.99875	0.58375	0.9925	-0.006249999999999989	
F1	0.9989010989010989	0.7356662460915538	0.9936284116750611	-0.005272687226037698	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9100000000000001	0.5999999999999999	0.8662500000000002	-0.04374999999999999	
F1	0.92384446063128	0.7451188521651313	0.8915114479500105	-0.032333012681269535	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6125	0.58375	0.6537499999999999	0.04125000000000001	
F1	0.5885198185573248	0.7282460980339465	0.7347757116631517	0.1462558931058268	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9525	0.5	0.8125	-0.13999999999999996	*
F1	0.9587663012688215	0.6562885179466105	0.8096347617854291	-0.14913153948339244	*

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.97875	0.56125	0.9724999999999999	-0.006250000000000033	
F1	0.9805471229310466	0.6950949729070162	0.9759354375466212	-0.004611685384425512	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.

ACCURACY	0.9537500000000001	0.5449999999999999	0.9087499999999998	-0.045000000000000005	
F1	0.9574646620033119	0.6592488663652769	0.9201864979704768	-0.03727816403283522	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9887500000000001	0.5637500000000001	0.9825000000000002	-0.006249999999999965	
F1	0.9903119429590017	0.7093553947567633	0.9849932032831987	-0.005318739675803208	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6637500000000001	0.4899999999999994	0.5737500000000001	-0.09	
F1	0.6377190651635288	0.5329450735770329	0.5456825583847361	-0.09203650677879277	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9825000000000002	0.5750000000000001	0.96875	-0.013750000000000007	
F1	0.9848619694663338	0.7124541025800035	0.9724332750742013	-0.012428694392132933	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6024999999999999	0.59875	0.6024999999999999	0	
F1	0.7507214882964476	0.7476243144675665	0.7507214882964476	0	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	48.0	80.0	1.02	1.0	0.0	nan	0.0	0.0	0.0	121.0
2	7.0	50.0	1.02	4.0	0.0	nan	0.0	0.0	0.0	nan
3	62.0	80.0	1.01	2.0	3.0	0.0	0.0	0.0	0.0	423.0
4	48.0	70.0	1.005	4.0	0.0	0.0	1.0	1.0	0.0	117.0
5	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
6	60.0	90.0	1.015	3.0	0.0	nan	nan	0.0	0.0	74.0
7	68.0	70.0	1.01	0.0	0.0	nan	0.0	0.0	0.0	100.0
8	24.0	nan	1.015	2.0	4.0	0.0	1.0	0.0	0.0	410.0
9	52.0	100.0	1.015	3.0	0.0	0.0	1.0	1.0	0.0	138.0
10	53.0	90.0	1.02	2.0	0.0	1.0	1.0	1.0	0.0	70.0
11	50.0	60.0	1.01	2.0	4.0	nan	1.0	1.0	0.0	490.0
12	63.0	70.0	1.01	3.0	0.0	1.0	1.0	1.0	0.0	380.0
13	68.0	70.0	1.015	3.0	1.0	nan	0.0	1.0	0.0	208.0
14	68.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	98.0
15	68.0	80.0	1.01	3.0	2.0	0.0	1.0	1.0	1.0	157.0
16	40.0	80.0	1.015	3.0	0.0	nan	0.0	0.0	0.0	76.0
17	47.0	70.0	1.015	2.0	0.0	nan	0.0	0.0	0.0	99.0
18	47.0	80.0	nan	nan	nan	nan	nan	0.0	0.0	114.0
19	60.0	100.0	1.025	0.0	3.0	nan	0.0	0.0	0.0	263.0
20	62.0	60.0	1.015	1.0	0.0	nan	1.0	1.0	0.0	100.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	41.0	60.0	1.0288724050195122	3.0	0.0	0.0	0.0	0.0	0.0	105.0
2	84.0	80.0	1.0076367591366582	0.0	0.0	0.0	0.0	0.0	0.0	139.0
3	68.0	80.0	1.0079297522058448	2.0	0.0	0.0	1.0	0.0	0.0	108.0
4	65.0	80.0	1.0146681186205875	0.0	0.0	1.0	1.0	0.0	1.0	86.0
5	57.0	100.0	1.0170269689109102	3.0	0.0	0.0	0.0	1.0	1.0	123.0
6	94.0	60.0	1.0187850330600947	0.0	0.0	0.0	0.0	0.0	0.0	124.0
7	35.0	80.0	1.005	3.0	0.0	1.0	0.0	0.0	0.0	nan
8	30.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	96.0
9	52.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	99.0
10	65.0	80.0	1.015	0.0	0.0	nan	0.0	0.0	0.0	115.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	94.0	90.0	1.0169740149295334	0.0	4.0	1.0	0.0	0.0	0.0	149.0
2	85.0	80.0	1.0160833739941828	3.0	0.0	0.0	0.0	0.0	0.0	148.0
3	94.0	90.0	1.0174810093219768	0.0	0.0	0.0	0.0	0.0	0.0	379.0
4	76.0	60.0	1.0160953886359567	4.0	2.0	0.0	0.0	0.0	0.0	380.0
5	97.0	80.0	1.0158557334095304	3.0	0.0	0.0	0.0	0.0	0.0	122.0
6	67.0	100.0	1.0251791374926382	2.0	0.0	0.0	0.0	0.0	0.0	100.0
7	64.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	113.0
8	42.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	98.0
9	12.0	60.0	1.015	3.0	0.0	1.0	1.0	1.0	0.0	nan
10	23.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	111.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	66.0	90.0	1.02021260924356 33	3.0	0.0	0.0	1.0	0.0	0.0	81.0
2	73.0	80.0	1.01851309116457 12	0.0	0.0	0.0	0.0	0.0	0.0	77.0
3	61.0	100.0	1.01952465926010 77	0.0	5.0	0.0	0.0	1.0	0.0	184.0
4	7.0	70.0	1.01816939687606 54	0.0	0.0	0.0	0.0	0.0	0.0	79.0
5	79.0	60.0	1.02460842661787 74	0.0	4.0	0.0	0.0	0.0	0.0	128.0
6	55.0	70.0	1.01426373253421 96	3.0	0.0	0.0	1.0	0.0	0.0	525.0
7	34.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	121.0
8	33.0	60.0	nan	nan	nan	0.0	0.0	0.0	0.0	130.0
9	60.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	124.0
10	57.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	132.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	36.0	80.0	1.01642679495134 04	1.0	0.0	0.0	0.0	0.0	0.0	119.0
2	75.0	50.0	1.01267679947434	0.0	0.0	0.0	0.0	0.0	0.0	87.0
3	59.0	90.0	1.02079604415578 8	2.0	0.0	0.0	0.0	0.0	0.0	80.0
4	43.0	70.0	1.01976668029189 48	0.0	0.0	0.0	0.0	0.0	1.0	56.0
5	86.0	80.0	1.01106217954869 58	3.0	0.0	0.0	1.0	0.0	0.0	92.0
6	75.0	80.0	1.02531964020331 71	4.0	0.0	0.0	0.0	0.0	0.0	234.0
7	44.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0
8	55.0	90.0	1.01	2.0	1.0	1.0	1.0	0.0	0.0	273.0
9	45.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	360.0
10	72.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	308.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	63.0	70.0	1.013703042876677	0.0	0.0	0.0	0.0	0.0	1.0	112.0
2	22.0	80.0	1.024223877479404	4.0	1.0	0.0	0.0	0.0	0.0	102.0
3	59.0	80.0	1.009642099964939	4.0	0.0	0.0	0.0	0.0	0.0	270.0
4	1.0	60.0	1.019866299835224	0.0	0.0	0.0	1.0	1.0	0.0	200.0
5	70.0	80.0	1.010825391149877	0.0	0.0	0.0	0.0	0.0	0.0	79.0
6	48.0	60.0	1.003561064301484	4.0	0.0	0.0	1.0	0.0	0.0	86.0
7	41.0	80.0	1.015	1.0	4.0	1.0	0.0	0.0	0.0	210.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	87.0
9	26.0	70.0	1.015	0.0	4.0	nan	0.0	0.0	0.0	250.0
10	50.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	137.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	57.0	90.0	1.024765924301674	2.0	0.0	0.0	0.0	0.0	0.0	119.0
2	53.0	60.0	1.016807797125695	2.0	1.0	0.0	0.0	0.0	0.0	116.0
3	91.0	90.0	1.023602789449846	4.0	2.0	0.0	0.0	0.0	0.0	147.0
4	61.0	90.0	1.012657608230442	2.0	0.0	0.0	0.0	0.0	0.0	165.0
5	55.0	80.0	1.024704394393126	0.0	0.0	0.0	1.0	0.0	1.0	162.0
6	12.0	60.0	1.020338910813162	4.0	0.0	1.0	0.0	1.0	0.0	113.0
7	8.0	60.0	1.025	3.0	0.0	0.0	0.0	0.0	0.0	78.0
8	60.0	70.0	1.01	1.0	0.0	nan	0.0	0.0	0.0	109.0
9	nan	70.0	1.01	3.0	0.0	0.0	0.0	0.0	0.0	110.0
10	62.0	90.0	1.02	2.0	1.0	nan	0.0	0.0	0.0	169.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	38.0	70.0	1.01955030034577 64	0.0	0.0	0.0	1.0	0.0	0.0	247.0
2	31.0	100.0	1.01801913757724 16	0.0	0.0	0.0	0.0	0.0	0.0	471.0
3	44.0	70.0	1.00911817250452 86	1.0	0.0	0.0	0.0	1.0	0.0	95.0
4	3.0	100.0	1.02249314166676 77	0.0	0.0	0.0	0.0	0.0	0.0	138.0
5	37.0	70.0	1.02332671808970 9	0.0	0.0	0.0	0.0	0.0	0.0	143.0
6	25.0	80.0	1.02200632573133 52	0.0	0.0	0.0	0.0	0.0	0.0	76.0
7	nan	70.0	1.02	0.0	0.0	nan	nan	0.0	0.0	219.0
8	29.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	127.0
9	50.0	90.0	1.015	1.0	0.0	1.0	1.0	0.0	0.0	nan
10	65.0	70.0	1.02	1.0	0.0	1.0	1.0	0.0	0.0	139.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	42.0	60.0	1.01215267988797 16	3.0	0.0	0.0	1.0	0.0	0.0	335.0
2	65.0	70.0	1.02001637811312 17	3.0	0.0	0.0	1.0	0.0	0.0	155.0
3	67.0	100.0	1.01210906079562 2	3.0	1.0	0.0	1.0	0.0	0.0	138.0
4	16.0	70.0	1.01545043058123 55	1.0	1.0	0.0	0.0	0.0	0.0	97.0
5	67.0	70.0	1.01588296362398 03	0.0	0.0	0.0	0.0	0.0	0.0	81.0
6	34.0	80.0	1.02466381977915 38	0.0	0.0	0.0	0.0	0.0	0.0	222.0
7	53.0	100.0	1.01	1.0	3.0	1.0	0.0	0.0	0.0	213.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	nan
9	45.0	80.0	1.02	3.0	0.0	0.0	1.0	0.0	0.0	425.0
10	33.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	89.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	59.0	80.0	1.0108422888544526	0.0	0.0	0.0	1.0	0.0	0.0	125.0
2	61.0	80.0	1.0257849728583954	3.0	0.0	0.0	1.0	0.0	0.0	119.0
3	11.0	70.0	1.0109607135297096	2.0	0.0	0.0	0.0	0.0	0.0	157.0
4	20.0	90.0	1.0229921828620385	0.0	0.0	0.0	1.0	0.0	0.0	248.0
5	57.0	80.0	1.0148107044769397	0.0	1.0	0.0	0.0	0.0	0.0	116.0
6	38.0	80.0	1.0182680299441838	2.0	0.0	1.0	1.0	0.0	0.0	346.0
7	46.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	102.0
8	22.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	97.0
9	65.0	80.0	1.01	3.0	3.0	nan	0.0	0.0	0.0	294.0
10	23.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	69.0	80.0	1.0220040011159641	0.0	0.0	0.0	0.0	0.0	0.0	95.0
2	55.0	70.0	1.0088414461899955	3.0	0.0	0.0	1.0	1.0	0.0	132.0
3	71.0	70.0	1.0187868482986335	0.0	0.0	0.0	0.0	1.0	0.0	186.0
4	56.0	80.0	1.0233610738716577	1.0	1.0	0.0	1.0	0.0	0.0	170.0
5	75.0	70.0	1.0184502346444755	0.0	0.0	0.0	1.0	1.0	0.0	547.0
6	88.0	80.0	1.0109370949871286	3.0	0.0	0.0	0.0	0.0	0.0	147.0
7	11.0	80.0	1.01	3.0	0.0	nan	0.0	0.0	0.0	nan
8	61.0	90.0	1.015	0.0	2.0	nan	0.0	0.0	0.0	nan
9	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
10	70.0	60.0	nan	nan	nan	nan	nan	0.0	0.0	120.0